

Muyao Kong

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EDUCATION

New York University

Major in Computer Science

Minor in Mathematics

Honors: Dean's List

Relevant Coursework: Natural Language Processing, Probability & Statistics, Linear Algebra, Basic Algorithm, Discrete Math, Data Structures

High School Attached to Shandong Normal University

GPA: 4.01 Ranking: 5%

New York, NY

Expected May 2027

Jinan, China

Graduated June 2023

RESEARCH

Read the Room Before Your Model Generates

Project Lead and Researcher

- Proposed **Scenario-Actor**, a framework that selects LLM decoding temperature from the communicative scenario before generation.
- Designed **Scenarize** to map r -th layer Transformer hidden states into an interpretable **scenario embedding** representing conversational context.
- Designed **Actout**, a scenario-axis readout function that maps scenario embeddings to scalar decoding temperatures.
- Designed a **triplet-supervised training** setup to learn scenario embeddings from relative differences among communicative contexts.

CVTime: Do LLMs Understand Chinese Vague Time Expressions in Context?

Researcher

- Introduced **CVTime**, the **first** systematic benchmark for evaluating LLM understanding of modern, cultural, and classical Chinese vague duration expressions.
- Constructed **252 base items** and **756 prompts** covering default interpretation, contextual adjustment, fictional-rule following, source-grounded reasoning, and relative comparison.
- Collected **3,024 responses** from 21 annotators and preserved full response distributions to capture variation in human judgments rather than using a fixed time band.
- Evaluated **six LLMs** with four main metrics - human-LLM agreement, ordinal distance, context-shift agreement, rule accuracy - and robustness analysis with three prompt variants and different answer orders.
- Found **limited human-LLM agreement** and **scaling effect in model size** within both Qwen and Llama families, and no reliable improvement from natural context.

Generate-then-Rerank Under Subjective Labels: Decomposing Generation and Selection Levers for Grade-Calibrated LLM Explanations

Researcher

- Developed a **generate-then-rerank** framework for subjective perceived-grade matching, separating generation and selection effects across Elementary, High School, and Graduate explanations.
- Built a fine-tuned **DistilBERT** perceived-grade reranker, outperforming the strongest feature-based scorer by 7 to 13 percentage points in paired rank accuracy within each question.
- Developed a diagnostic using mean pool quality and within-cell correlation to identify generation- and selection-side bottlenecks and quantify a **14.1-point oracle gap**.
- Conducted three human-annotation studies with over 30,000 votes, finding the best measured sampling temperature at **T=1.5** and a gain of **6.2 points** on deployment share from reranking.

PUBLICATION & MANUSCRIPT

[1] **Muyao Kong**, and Zhihao Lin. "Read the Room Before Your Model Generates." *Under review*.

[2] **Muyao Kong**. "CVTime: Do LLMs Understand Chinese Vague Time Expressions in Context?" 2026 (manuscript in preparation).

[3] Muqiao Tao, **Muyao Kong**, and Qitian Xiong. "Generate-then-Rerank Under Subjective Labels: Decomposing Generation and Selection Levers for Grade-Calibrated LLM Explanation." *Under review*.

INDUSTRIAL EXPERIENCE

Alibaba Group

Artificial Intelligence Intern

Hangzhou, China
Jun 2024 - Jul 2024

- Deployed local SD1.5 pipeline and built full-stack web system for avatar generation. Constructed mapping rules from structured parameters to diffusion prompts, and **benchmarked** performance and hardware requirements of mainstream generative models including Stable Diffusion and Flux
- Leveraged Hy-MT2-7B and Qwen3-4B models to optimize prompt engineering. Designed **incremental semantic merging strategies** to resolve semantic conflicts and redundant descriptions in multi-round generation, and summarized stable model tuning solutions through comparative tests of CFG scale, sampling steps and seed parameters
- Implemented **T2I** and **img2img** generation pipelines, and adapted dual inference backends including local Stable Diffusion and ComfyUI. Designed interface fields for historical prompts, incremental instructions and denoising strength to support end-to-end iterative image editing with subject consistency
- Optimized denoising strength thresholds for img2img iteration. Established classification rules for instruction addition, replacement and deletion to mitigate feature drift and semantic conflicts, improving the stability and accuracy of iterative image modification
- Conducted **full-scene testing** and **iterative optimization** of multimodal generation system. Built prompt conflict resolution mechanisms based on hard constraints and user latest intentions, and sorted out test cases and regression documents to enhance system stability and maintainability

National Supercomputing Center

Algorithm and Data Analyst Intern

Jinan, China
Jun 2024 - Aug 2024

- **Cleaned** time series device data; constructed data anomaly detection and localization algorithms to locate and process abnormal data data using **Huber Regressor** in **robust regression**.
- Built **ICC (Intraclass Correlation)** data consistency check algorithms, made judgment of consistency on the historical data of multiple related devices.
- Built machine learning **ARIMA** (Autoregressive Integrated Moving Average) model, used the constructed ARIMA model to forecast the **time series** device data.
- Wrote Python automation script for timed processing of device data, handled timed database read/write using **PyMySQL**, and processed the retrieved device data from MySQL database.

China Cinda Asset Management Co., Ltd

Risk Management Intern

Beijing
Jun 2025 – Aug 2025

- Synthesized industry and internal case materials on **distressed asset resolution** (e.g., **debt-to-equity swaps, auctions, restructurings**) and distilled key risk factors into reusable briefings for internal reference.
- Compiled and standardized annual report data across 18 financial institutions including banks, AMCs and brokers, extracting and consolidating key financial and disclosure items into structured datasets for analysis.
- Performed basic **cross-company** and **time-series comparisons** (leverage, liquidity, profitability quality, impairments, contingencies) and drafted concise risk highlights to support internal documentation.
- Operated under strict data-access and compliance constraints, focusing on public disclosures and documentation workflows to improve traceability and retrieval efficiency.

TECHNICAL SKILLS

Programming Languages: Proficient - Python, Java, C++, JavaScript, Familiar - C
Markup Languages: Proficient - LaTeX, HTML, CSS
Other Libraries: Proficient - Pandas, NumPy, Matplotlib, Familiar - Statsmodel, PyTorch

OTHER SKILLS

Languages: Mandarin, English, Japanese, Arabic
Interests: Artificial Intelligence, Mathematics, Foreign Languages, Linguistics, Economics, Designing, Piano